

PARAMEDIC PHARMACOLOGY CHEAT SHEET

18 Critical EMS Medications

Drug	Class	Indication	Adult Dose	Peds Dose	Route	Contra/SE
Epinephrine	Sympathomimetic	Cardiac arrest, Anaphylaxis, Severe asthma	1mg IV/IO (1:10,000) q3-5min	0.01mg/kg IV/IO (1:10,000)	IV/IO	HTN, Tachycardia, Dysrhythmias
Amiodarone	Antiarrhythmic	VF/pVT, Stable VT	300mg IV/IO push, then 150mg	5mg/kg IV/IO	IV/IO	Hypotension, Bradycardia, Heart block
Lidocaine	Antiarrhythmic	VF/pVT (alternative to amiodarone)	1-1.5mg/kg IV/IO, repeat 0.5-0.75mg/kg	1mg/kg IV/IO	IV/IO	Seizures, Bradycardia, Heart block
Atropine	Anticholinergic	Symptomatic bradycardia	0.5mg IV q3-5min (max 3mg)	0.02mg/kg IV (min 0.1mg)	IV/IO	Tachycardia, Glaucoma, Hypothermia
Adenosine	Antiarrhythmic	Stable SVT	6mg rapid IV push, then 12mg if needed	0.1mg/kg (max 6mg), then 0.2mg/kg if needed	IV/IO	Bradycardia, Spasm, Transient asystole
Dopamine	Sympathomimetic	Symptomatic bradycardia (2nd line)	2-20mcg/kg/min IV drip	2-20 mcg/kg/min IV drip	IV drip	Tachydysrhythmias, HTN
Naloxone (Narcan)	Opioid antagonist	Opioid overdose	0.4-2mg IV/IO/IM/IN	0.1mg/kg IV/IO/IM/IN (max 2mg)	IV/IO/IM/IN	Opioid withdrawal, Pulmonary edema
Dextrose (D50/D25/D10)	Carbohydrate	Hypoglycemia	25g IV (D50W 50mL)	0.5-1g/kg IV (D25W 2-4mL/kg, D10W 5-10mL/kg)	IV	Hypertonicity, Tissue necrosis
Midazolam (Versed)	Benzodiazepine	Seizures, Sedation, Cardioversion	2-5mg IV/IO/IM/IN	0.1mg/kg IV/IO/IM/IN (max 5mg)	IV/IO/IM/IN	Respiratory depression, Hypotension
Diazepam (Valium)	Benzodiazepine	Seizures, Anxiety, Muscle spasm	5-10mg IV/IO	0.1-0.3mg/kg IV/IO	IV/IO/IM	Respiratory depression, Hypotension
Morphine Sulfate	Opioid analgesic	Pain management, ACS, Pulmonary edema	2-4mg IV q5min PRN	0.1-0.2mg/kg IV	IV/IO/IM/SQ	Respiratory depression, Hypotension
Fentanyl	Opioid analgesic	Pain management (preferred over morphine)	50-100mcg IV/IO/IM/IN	1-2mcg/kg IV/IO/IN	IV/IO/IM/IN	Respiratory depression, Chest wall rigidity
Nitroglycerin	Vasodilator	ACS, CHF, Hypertensive emergency	0.4mg SL q3-5min (max 3 doses)	Not typically used peds	SL	Hypotension, PDE5 inhibitor use, Head injury
Aspirin	Antiplatelet	Suspected ACS	324mg chewed (non-enteric)	Not typically used peds	PO (chewed)	Allergy, Active bleeding, Children (Reye syndrome)
Albuterol	Beta-2 agonist	Bronchospasm, Hyperkalemia	2.5-5mg nebulized	2.5mg nebulized (<20kg), 5mg (≥20kg)	IV/IO/IM	Tachycardia, Dysrhythmias
Diphenhydramine (Benadryl)	H1 histamine	Allergic reaction, Dystonic reaction	25-50mg IV/IO/IM	1mg/kg IV/IO/IM (max 50mg)	IV/IO/IM	Glaucoma, Urinary retention, Sedation
Glucagon	Hormone	Hypoglycemia (no IV access), Beta-blocker (no hypoglycemia)	1-3mg IM/IO, 3-10mg IV	0.5mg IM/IO (<20kg), 1mg IM/IO (≥20kg)	IM/IO	Nausea, Vomiting
Sodium Bicarbonate	Alkalizing agent	Tricyclic OD, Hyperkalemia, Metabolic acidosis	1mEq/kg IV	1mEq/kg IV	IV	Metabolic alkalosis, Hypernatremia

Clinical Pearls

- Epinephrine 1:10,000 = 0.1mg/mL (for IV/IO use in cardiac arrest)
- Epinephrine 1:1,000 = 1mg/mL (for IM use in anaphylaxis)
- Always confirm D50 concentration before administering to pediatrics
- Naloxone duration of action (30-90 min) may be shorter than opioid - monitor for re-narcotization
- Nitroglycerin is CONTRAINDICATED with PDE5 inhibitor use (Viagra, Cialis) within 24-48 hours
- Fentanyl is preferred over morphine in hypotensive patients due to less histamine release
- Atropine minimum single dose is 0.1mg to avoid paradoxical bradycardia
- Adenosine requires RAPID push followed by 20mL saline flush

DISCLAIMER: This product contains general medical reference information for educational purposes only. It does not constitute medical advice, clinical guidance, or protocol. Always follow your local protocols, medical direction, and institutional guidelines. Clinical accuracy should be verified against current evidence-based resources before use in patient care.